HZO based FeFET with Sub 2-nm EOT Gate Stack as Synapse for Spiking Neural Network

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***Abstract*—Ferroelectric FET (FeFET) synapses based on HZO with a sub-2 nm EOT gate stack promise dense, low-power neuromorphic hardware. Yet, their system efficacy is tightly coupled to the particulars of device-level electrical characterization. In this work, we establish a quantitative device-to-algorithm link by evaluating how standard measurement controls such as program/erase pulse amplitude, pulse duration, and cycling endurance, shape the FeFET memory window, weight uniformity and, in turn, impact spiking neural network (SNN) performance. Using experimentally measured transfer characteristics and pulse responses, we show that reductions in memory window and increased update non-uniformity induced by sub-optimal pulse conditions directly constrain the attainable weight range and weight-change linearity, degrading accuracy and stability during SNN training/inference. We map the feasible operating envelope that jointly maximizes memory window, monotonicity, and endurance, and translate it into actionable pulse-engineering guidelines (amplitude–width co-tuning and endurance-aware programming schedules) for synaptic updates. By closing the loop from metrology to application, this study provides a systematic methodology to co-optimize FeFET electrical characterization and SNN performance, offering practical screening criteria for wafer-level device selection and a pathway to reproducible, deployment-grade neuromorphic systems with sub-2 nm-stack FeFET synapses.**

***Index Terms*—FeFET, Spiking Neural Network, Synapse, Non-volatile Memory, Neuromorphic Computing**

I. Introduction

The "memory wall" in von Neumann architectures has intensified, motivating compute paradigms that collocate memory and processing. Neuromorphic approaches inspired by the brain offer compelling gains in area and energy efficiency [1], [2]. Among these, SNNs aim to exploit sparse, event-driven computation in hardware; however, CMOS-only realizations struggle to meet brain-like efficiency at practical scale [3]–[5]. To bridge this gap, device communities have pursued emerging memories such as RRAM, MRAM, phase-change memory, and ferroelectric memories, as compact, low-energy synaptic elements [6]–[9].

Within this landscape, FeFETs are especially attractive: non-volatile polarization switching, low-voltage operation, fast updates, and CMOS compatibility enable dense arrays with analog, gradual conductance modulation. The resulting multi-level update behavior aligns naturally with synaptic weight storage and in-situ learning. Here, we demonstrate FeFET synapses comprising an HZO layer of ∼5 nm embedded in a sub-2 nm EOT gate stack and evaluate end-to-end SNN performance on MNIST. We systematically vary programming conditions such as, pulse amplitude, pulse width, and cycling endurance, to quantify how these knobs shape the memory window, weight-update linearity/monotonicity, and ultimately classification accuracy. This device-to-algorithm analysis identifies operating regimes that preserve a wide, uniform update range while sustaining endurance, providing practical guidance for pulse engineering and screening of wafer-level devices destined for neuromorphic systems.

II. Experimental Details

Solid–solution hafnia–zirconia (SS–HZO, $Hf_{0.5}Zr_{0.5}O_2$) gate dielectrics were deposited on p-Si/SiO_2 by thermal ALD using alternating HfO_2 and ZrO_2 cycles. A balanced super-cycle maintained an Hf:Zr cation ratio of 1:1, yielding a thickness of ∼5 nm. Devices were realized in a gate-first flow with a TaN/SS–HZO/SiO_2/p-Si stack. Post-deposition rapid thermal annealing (Ar ambient) was performed to crystallize the SS–HZO and activate dopants. The full process sequence and the 3D device schematic are shown in Fig. 1.

Figure 2(a) maps the extracted V_{th} response across a systematic amplitude–width sweep and reveals a reproducible, analog-like update behavior, that is, for each fixed pulse width, the threshold first shifts in the potentiation direction and then

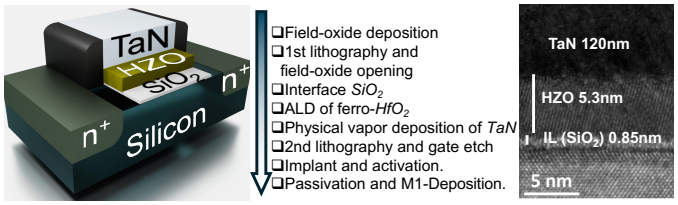
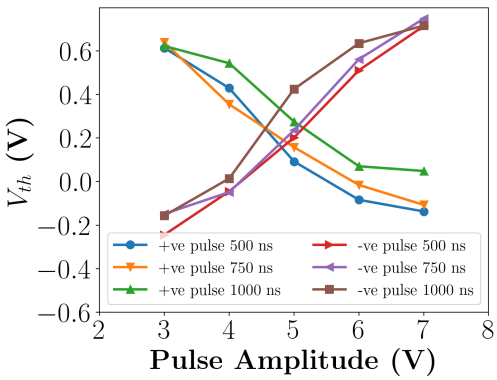
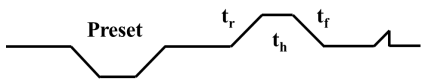


Fig. 1: **Schematic and process-flow for the fabricated FeFET**: The gate stack comprises TaN (120 nm) / HZO (∼5.3 nm) / interfacial SiO_2 (IL, ∼0.85 nm) on Si. The flow (right) illustrates: field-oxide deposition and lithographic opening; IL formation; ALD of ferroelectric HfO_2 (HZO); physical-vapor deposition of TaN; second lithography and gate etch; source/drain implant & activation; and final passivation with M1 metallization.



(a)



(b)

Fig. 2: **Threshold Voltage Control:** (a) Measured threshold voltage shift (V_{th}) versus programming pulse amplitude for both positive (potentiation) and negative (depression) pulses at three pulse widths: 500 ns, 750 ns, and 1000 ns. Pulse amplitude was swept from ±3 V to ±7 V, with 1 V increments/decrements. At a given width, the device exhibits an increase in V_{th} over a portion of the amplitude sweep, followed by a decrease at larger amplitudes. (b) Pulse timing used for the measurements: a 7 V preset (100 ns) initializes the device, then the program pulse with variable hold time t_h is applied; rise and fall times (t_r, t_f) were fixed at 20 ns for all experiments so that the effective pulse width was controlled solely via t_h.

toward depression as the drive is increased. This controllable modulation of V_{th} demonstrates that the device can implement multi-level weight updates. However, it exposes an operational trade-off: longer hold times produce larger, smoother, and more monotonic shifts (beneficial for per-pulse linearity). In contrast, shorter holds give finer per-pulse granularity at the expense of monotonicity and cause the device to saturate faster near the memory window edges. The pulse timing used in these measurements is described in Fig. 2(b). The data indicate that co-tuning amplitude and hold time (favoring mid-range amplitudes with moderate hold durations) provides optimization between update size, linearity, and endurance for neuromorphic computing applications.

III. Spiking Neural Network

A two-layer customized spiking neural network (SNN) is utilized on the MNIST dataset to evaluate the FeFET as synapse [10]. First, the experimental data (taken at room temperature) is processed by scaling and normalizing it. Next, a fixed number of states each FET will have is evaluated to understand the requirement on the number of states between a given memory window on an accuracy metric. Finally, the

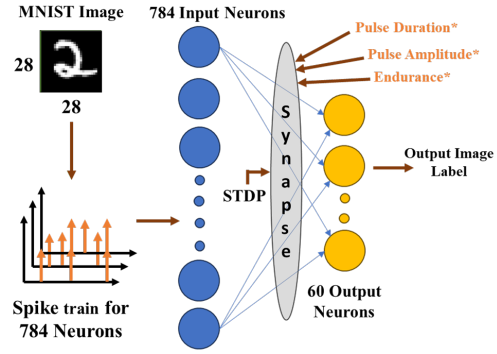


Fig. 3: **Overview of the SNN architecture**: Each 28 × 28 MNIST image is converted into a spike train, generating 784 spike-encoded inputs that stimulate the 784 input neurons. These inputs are connected through FeFET synapses to 60 output neurons in a fully connected configuration. Synaptic weight updates follow the standard STDP learning rule. The parameters under study, pulse duration, pulse amplitude, and endurance (*), govern the update dynamics of the FeFET synapses.

FeFET device is compared against an ideal device by considering the FeFET both with and without endurance effects (repeated pulses cause a decrease in $V_{th,max} - V_{th,min}$) as synapses in the SNN. Fig. 3 gives an overview of the SNN we have used for evaluation.

A. FeFET Data Preprocessing and State Count Determination

The V_{th} values for the given data are interpolated and normalized to be between 0 and 1. Assuming an infinite number of states and no endurance effect on the FET, the ideal accuracies are evaluated on the 5-class MNIST dataset. Subsequently, we discretize the number of states to 100, 50, 42, and 25 (with pulse width of 1000 ns) to understand and determine a representative number of states.

B. Network Architecture and Neurons

A two-layer spiking neural network with 784×60 neurons for a classification task between digits 0, 1, 2, 3, 4 is shown in Fig. 3. All input neurons are connected through synapses with all output neurons, as it is a fully connected network. Here, FeFET is used as a synapse to evaluate its weight change efficacy for different pulse amplitude/durations along with endurance for its classification accuracy in a pattern recognition task such as 5-class MNIST.

The neuron model used is Leaky Integrate and Fire (LIF) neurons with adaptive thresholds to prevent repetitive consecutive firing of output neurons [10]. In hardware, this is implemented as a resistor-capacitor circuit, with a voltage comparator, where the threshold voltage itself varies around V_{th}, based on the frequency of the spiking neuron.

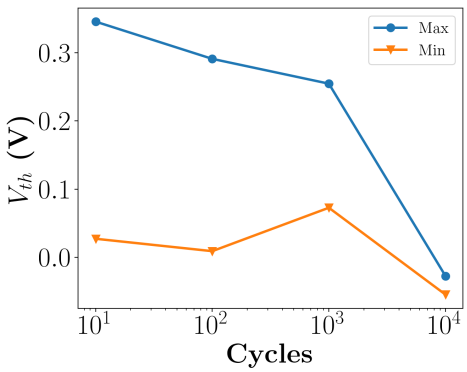


Fig. 4: **Threshold voltage limit with repeated pulses**: As the number of update cycles grows, the maximum and minimum V_{th} values gradually converge, shrinking the available memory window. This shows how endurance stress reduces the usable range for synaptic weight updates in FeFET devices.

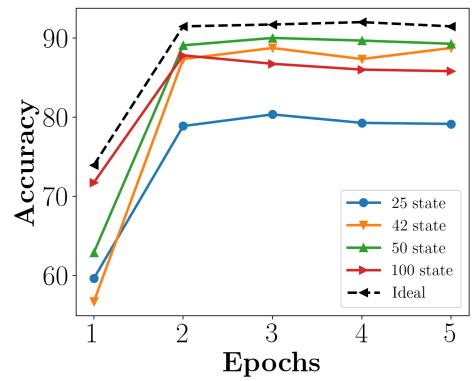


Fig. 5: **Classification accuracy** over five epochs for synapses with different discretized state counts. Increasing the number of states brings the device behavior closer to an ideal infinite-state synapse, improving accuracy. Since a real FeFET supports only a finite number of stable states, the 42-state case is adopted as a representative operating point.

C. Learning Rule and Input Data Encoding

Learning rule defines how the synapse weight should change, based on the time difference between the spiking neurons. We use the standard STDP learning rule, as follows:

$$\Delta w = \begin{cases} A_{up} \cdot e^{-\frac{\Delta t}{\tau_{up}}}, & \Delta t \geq 0 \\ A_{down} \cdot e^{-\frac{\Delta t}{\tau_{down}}}, & \Delta t < 0 \end{cases} \quad (1)$$

Where $A_{up} = 0.8$, $A_{down} = -1.3$, $\tau_{up} = \tau_{down} = 5$.

The data is passed through the network as spike trains for a fixed duration of 100 timesteps. Each image is 28×28 pixels, having an intensity value between 0 and 255. This is mapped linearly to a frequency between 5 Hz and 70 Hz. Hence we get a binary matrix $A \in \mathbb{R}^{784\times100}$, where each column is a spike train, and $A[i][j] = 1$ if i^{th} neuron fired at j^{th} timestep, and 0 otherwise.

D. FeFET Endurance

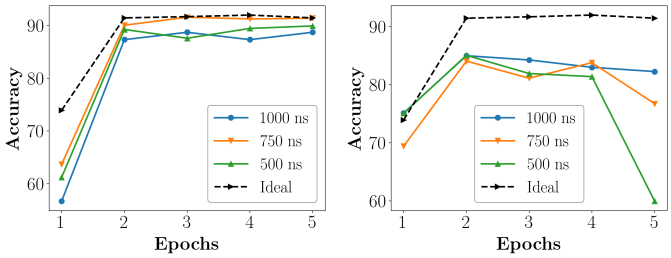
To understand the impact of repetitive update on the FeFET memory window, the cyclic endurance measurements are performed. The V_{th} update of the FeFET with the $V_{th,max}$ and $V_{th,min}$ are shown in Fig. 4. As the number of cycles moves closer to 10^4, the memory window collapses giving minimal margin to differentiate between high/low weight.

To incorporate endurance in the SNN, each synapse is tracked for number of weight changes during learning, and map the updated weight to the corresponding weight range using endurance data.

IV. Results and Discussion

A. Finding a Representative State Count

SNN performance is subsequently evaluated for modeling different weights (from high till FeFET states) to understand the requirement on the weight change uniformity and numbers. Testing the model on 100, 50, 42, and 25 states provides the accuracies shown in Fig. 5. It can be observed that a low state count (25) significantly degrades accuracy, while performance saturates for higher counts (42, 50, and 100), with only marginal differences between them. To evaluate further performance, a 42-state synapse is chosen, as it balances the number of states, the loss in accuracy against the ideal synapse, and hardware realism [11].



(a) (b)

Fig. 6: **Effect of programming pulse duration on classification accuracy** over five training epochs. (a) **Varying pulse duration without endurance:** When endurance is not modeled, a mid-range pulse (750 ns) reaches near-ideal accuracy fastest, while the shortest pulse (500 ns) lags. (b) **Varying pulse duration with endurance:** When endurance effects are included, accuracy deteriorates with epoch count, and the shortest pulse (500 ns) shows the largest long-term drop; longer pulses (1000 ns) retain performance more steadily.

B. Variation in Pulse Width

By keeping the pulse amplitude constant (at 5V), and varying the pulse width between 1000 ns, 750 ns, and 500 ns without considering the endurance-based performance, the accuracies can be observed in Fig. 6(a) with 750 ns duration demonstrating better accuracy.

When endurance is included in the simulation (Fig. 6(b)), a steady drop in accuracy across epochs driven by a gradual reduction in the device memory window is observed. Endurance reduces $V_{th,max} - V_{th,min}$ and make per-pulse weight updates

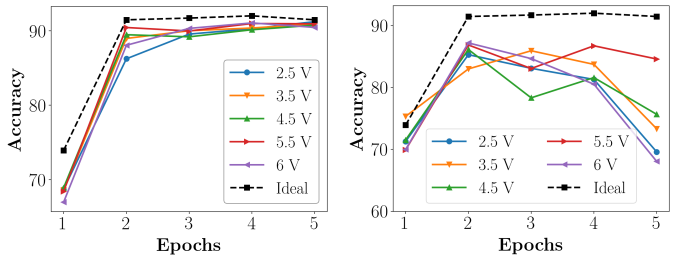


Fig. 7: **Effect of programming pulse amplitude on classification accuracy** over five training epochs. (a) **Varying pulse amplitude without endurance:** Without endurance modeling, accuracy converges rapidly for most amplitudes tested. (b) **Varying pulse amplitude with endurance:** When endurance is included, amplitude-dependent degradation appears; some amplitudes retain accuracy across epochs better than others.

more nonuniform overlapping the previously distinguishable states. The net result is fewer reliable levels for synaptic encoding and degradation of classification accuracy.

C. Variation in Pulse Amplitude

By keeping the pulse width constant at 500 ns, and varying the pulse amplitude from 2.5 V to 6 V, we observe the accuracies shown in Fig. 7(a) without endurance consideration. No drastic difference has been observed due to minimal variation in the memory window.

When endurance is included (Fig. 7(b)), the choice of pulse amplitude changes both the initial memory-window margin and how quickly that margin erodes under repeated pulses, producing the observed amplitude-dependent accuracy trends. In our experiments (500 ns width), amplitudes in the mid-to-high range (around 5–5.5 V) give the best trade-off between accuracy and endurance robustness.

V. Conclusion

We established a quantitative device–algorithm linkage for HZO-based FeFET synapses on MNIST, showing that *endurance* is the dominant limiter: cycling narrows the effective memory window $W \triangleq V_{\mathrm{th,max}} - V_{\mathrm{th,min}}$ and elevates cycle-to-cycle and device-to-device variances $(\sigma_{\mathrm{c2c}}, \sigma_{\mathrm{d2d}})$, which together depress accuracy. Programming conditions (amplitude/width) that further compress W or induce uneven per-pulse ΔV_{th}, captured by weight-update nonlinearity η and potentiation/depression asymmetry A, exacerbate this loss by clipping usable weight range and destabilizing convergence. Empirically, accuracy scales monotonically with W and inversely with $\{\sigma_{\mathrm{c2c}}, \sigma_{\mathrm{d2d}}, \eta, A\}$; preserving the effective analog resolution under cycling is more valuable than maximizing fresh W. These results translate into deployment specifications: maintain a wide, stable W across cycling/reads; ensure monotonic, near-linear, and balanced updates (low η, low A); bound stochasticity (low σ_{c2c}) and lot/device spread (low σ_{d2d}); and suppress drift between inference intervals. Endurance-aware pulse engineering (amplitude–width co-tuning, verify-gated burst updates, limited overdrive) combined with process control of HZO composition/thickness, interfacial layer/electrodes, and anneal budget curbs variability. A wafer-level screening protocol reporting $\{W, \eta, A, \sigma_{\mathrm{c2c}}, \sigma_{\mathrm{d2d}}, S_{\mathrm{eff}}\}$ pre/post-endurance, with device-in-the-loop task projections, closes the loop from metrology to application and enables actionable acceptance criteria and process/pulse co-optimization for reproducible, deployment-grade neuromorphic systems.

VI. Acknowledgement

This work was supported by NSTC-Taiwan (NSTC 114-2222-E-007-001, 114-2811-E-007-001), TSRI (JDP114-Y1-002), PSMC (PSMC-114A0), the Anusandhan National Research Foundation (ANRF/ECRG/2024/000901/ENS), and through the tools made available under the C2S project of the MEITY, Government of India.

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